3.2 Learning assumptions of the forward process

We consider the sequence of densities p_n in the forward process in (19). Recall from Section 2.3 that for fixed step-size $\gamma > 0$, the n-th step classical JKO scheme finds p_{n+1} by minimizing

$$\min_{\rho \in \mathcal{P}_2} F_{n+1}(\rho) := G(\rho) + \frac{1}{2\gamma} \mathcal{W}_2^2(p_n, \rho), \tag{24}$$

where $G(\rho) = \mathrm{KL}(\rho || q)$. The learning in the n-th Residual Block in a JKO flow network computes the minimization via parameterizing the transport T_{n+1}. Here we briefly review the rational of solving (24) by solving for T_{n+1}, which leads to our assumption of the learned forward process.

JKO step by learning the transport. In the right hand side of (24), when both p_n and ρ are in $\mathcal{P}_2^r$, the Brenier Theorem (Theorem 2.1) implies the existence of a unique OT map T from p_n to ρ. Consider the following minimization over the transport T,

$$\min_{T:\mathbb{R}^d \to \mathbb{R}^d} G(T_\# p_n) + \frac{1}{2\gamma} \mathbb{E}_{x \sim p_n} \|x - T(x)\|^2 . \tag{25}$$

The following lemma, proved in Appendix A, shows that the minimizer T makes $T_\# p_n \in \mathcal{P}_2^r$:

Lemma 3.3. *Suppose $p_n \in \mathcal{P}_2^r$ makes $G(p_n) < \infty$, and T is a minimizer of (25), then $T_\# p_n \in \mathcal{P}_2^r$.*

Thus, in (25) it is equivalent to minimize over T that renders $T_\# p_n \in \mathcal{P}_2^r$ and this means that the minimizer T is the OT map. One can also verify that (25) is equivalent to (24) in the sense that a minimizer T^* of (25) makes $(T^*)_\# p_n$ a minimizer of (24), and for a minimizer ρ^* of (24) the OT map from p_n to ρ^* is a minimizer of (25) [74, Lemma A.1].

Learning error in JKO flow network. In the n-step of the JKO flow network, the transport T in (25) is parameterized by a Residual block, and the learning cannot find the T that exactly minimizes (25) (and equivalently (24)) for three reasons:

(i) Approximation error: The minimization of (25) is over T constrained inside some neural network family $\mathcal{T}_\Theta$. When the function family $\mathcal{T}_\Theta$ is large enough to express the desired optimal transport from p_n to $\mathrm{Prox}_{\gamma G}(p_n)$, the solution can approximate the exact minimizer of (24), but this usually cannot be guaranteed.

(ii) Finite-sample effect: The training is computed on empirical data samples, while in this analysis, we focus on the minimization of population loss.

(iii) Imperfect optimization: The learning of neural networks is a non-convex optimization typically implemented by Stochastic Gradient Descent (SGD) over mini-batches and there is no guarantee of achieving a minimizer of the empirical loss.

As a result, the learned transport T_{n+1} finds a $p_{n+1} = (T_{n+1})_\# p_n$ that at most approximately minimizes (24). While the learned T_{n+1} is usually not the exact minimizer, we assume that it is regular enough such that p_{n+1} is still in $\mathcal{P}_2^r$. This would hold if T_{n+1} is non-degenerate and also in $L^2(p_n)$ by the following lemma proved in Appendix A.

Lemma 3.4. *Let $p \in \mathcal{P}_2^r$, then $\{T : \mathbb{R}^d \to \mathbb{R}^d,\ T \in L^2(p)\} = \{T : \mathbb{R}^d \to \mathbb{R}^d,\ T_\# p \in \mathcal{P}_2\}$. As a result, if $T \in L^2(p)$ and is non-degenerate, then $T_\# p \in \mathcal{P}_2^r$.*

When p_n and p_{n+1} are both in $\mathcal{P}_2^r$, we have a unique invertible OT map from p_n to p_{n+1} by Brenier Theorem, and its (a.e.) inverse is the OT map from p_{n+1} to p_n. Specifically, we define

$$\begin{aligned} &T_n^{n+1} \text{ is the OT map from } p_n \text{ to } p_{n+1}, \quad p_n\text{-a.e.,} \\ &T_{n+1}^n \text{ is the OT map from } p_{n+1} \text{ to } p_n, \quad p_{n+1}\text{-a.e.,} \end{aligned} \tag{26}$$

and we have $T_{n+1}^n \circ T_n^{n+1} = \mathrm{I_d}$ p_n-a.e., $T_n^{n+1} \circ T_{n+1}^n = \mathrm{I_d}$ p_{n+1}-a.e. Note that T_n^{n+1} differs from the learned map T_{n+1}, see Remark 3.2.

Assumption on approximate first-order condition. For our analysis, we theoretically characterize the error in learning T_{n+1} by quantifying the error in the first-order condition. Specifically, the $\mathcal{W}_2$-gradient (strictly speaking, a sub-differential) of F_{n+1} at ρ can be identified as

$$\nabla_{\mathcal{W}_2} F_{n+1}(\rho) = \nabla_{\mathcal{W}_2} G(\rho) - \frac{T_\rho^n - \mathrm{I_d}}{\gamma}, \quad \rho\text{-a.e.} \tag{27}$$

where T_ρ^n is the OT map from ρ to p_n. Here we use $\nabla_{\mathcal{W}_2}\phi$ to denote the sub-differential $\partial_{\mathcal{W}_2}\phi$ assuming unique existence to simplify exhibition. The formal statement in terms of subdifferential is provided in Lemma A.1 (which follows the argument of [7, Lemma 10.1.2]) for more general G (which includes KL divergence as a special case). For KL divergence G, if V is differentiable, the sub-differential $\partial_{\mathcal{W}_2} G$ is reduced to the unique $\mathcal{W}_2$-gradient written as

$$\nabla_{\mathcal{W}_2} G(\rho) = \nabla V + \nabla \log \rho, \tag{28}$$

when $\rho \in \mathcal{P}_2^r$ has a well-defined score function.

If ρ is the exact minimizer of (24), we will have $\nabla_{\mathcal{W}_2} F_{n+1}(\rho) = 0$ (and for sub-differential the condition is $0 \in \partial_{\mathcal{W}_2} F_{n+1}(\rho)$). At p_{n+1} which is pushed-forward by the learned T_{n+1}, we denote the $\mathcal{W}_2$-gradient of F_{n+1} by the following (recall the definition of T_{n+1}^n as in (26))

$$\begin{aligned} \xi_{n+1} &:= \nabla_{\mathcal{W}_2} F_{n+1}(p_{n+1}) \\ &= \nabla_{\mathcal{W}_2} G(p_{n+1}) - \frac{T_{n+1}^n - \mathrm{I_d}}{\gamma}, \quad p_{n+1}\text{-a.e.} \end{aligned} \tag{29}$$

and it is interpreted as sub-differential when needed. Making an analogy to the (approximate) first-order condition in vector-space optimization, we assume that p_{n+1} is close to the exact minimizer such that the (sub-)gradient ξ_{n+1} is small. In practice, the SGD algorithm to minimize the training objective (25) (assuming $\mathcal{T}_\Theta$ is expressive enough to approximate the exact minimizer T^*) would stop making progress when the $\mathcal{W}_2$-gradient vector field ξ_{n+1} evaluated on data samples collectively give a small magnitude. We characterize this by a small $L^2(p_{n+1})$ norm of the (sub-)gradient ξ_{n+1} in our theoretical assumption. The assumptions on the learned transport T_{n+1} are summarized as follows:

Assumption 1 (Approximate n-th step solution)**.** *The learned transport T_{n+1} is non-degenerate and in $L^2(p_n)$; it is invertible on $\mathbb{R}^d$ and T_{n+1}^{-1} is also non-degenerate. In addition, for some $\varepsilon > 0$, $\exists \xi_{n+1} \in \partial_{\mathcal{W}_2} F_{n+1}(p_{n+1})$ s.t.*

$$\|\xi_{n+1}\|_{p_{n+1}} \leq \varepsilon. \tag{30}$$

We experimentally verify the smallness of $\|\xi_{n+1}\|_{p_{n+1}}$ in neural network training of one JKO step. Figure A.1 shows the decrease of the squared L^2 norm $\|\xi_{n+1}\|_{p_{n+1}}^2$ over training iterations. The Wasserstein gradient ξ_{n+1} is plotted as a vector field in Figure A.2, whose magnitude gradually decreases over training. See Appendix D for more details.

The error magnitude ε can be viewed as an algorithmic parameter that reflects the accuracy of first-order methods, and similar assumptions have been made in the analysis of stochastic (noisy) gradient descent in vector space, see, e.g., [55, 56]. We emphasize that theoretically, ε does not need to be small but will enter the final error bound. For T_{n+1} satisfying Assumption 1, from $p_n \in \mathcal{P}_2^r$, $p_{n+1} = (T_{n+1})_\# p_n$ is also in $\mathcal{P}_2^r$ by Lemma 3.4. Then the subdifferential $\partial_{\mathcal{W}_2} F_{n+1}$ can be defined at p_{n+1} and characterized by Lemma A.1.

Remark 3.1 (Assumptions on T_n). The L^2 integrability condition of T_n is natural and together with non-degeneracy ensures that the next p_n is in $\mathcal{P}_2^r$. When T_n^{-1} is also non-degenerate, the p_n's in the forward process and q_n's in the reverse process in (19) all have densities, and this allows us to apply the data processing inequality in both directions (Lemma 5.1). When there is an inversion error in the reverse process, we will further impose Lipschitzness of T_n^{-1} on $\mathbb{R}^d$ and the Lipschitz constant will theoretically enter the error bound, see more in Section 5.2.

Remark 3.2 (T_{n+1} and T_n^{n+1}). Recall that T_{n+1} is the learned transport map and T_n^{n+1} is the OT map. In our setting (of imperfect minimization in the n-th step), both T_{n+1} and T_n^{n+1} push p_n to p_{n+1} but they are not necessarily the same. The notion of T_n^{n+1} is introduced only for theoretical purposes (the existence and invertibility are by Brenier Theorem), and our theory do not need T_{n+1} to equal T_n^{n+1}. On the other hand, one would expect T_{n+1} to approximate T_n^{n+1} when the JKO-step optimization (24) is approximately solved, leading to a small ε in (30).

3.3 Reverse process with inversion error

Considering potential inversion error in the reverse process, we denote the sequence of transports as S_n and the transported densities as $\tilde{q}_n$, that is,

$$\tilde{q}_n = (S_{n+1})_\# \tilde{q}_{n+1}, \tag{31}$$

from $\tilde{q}_N = q_N = q$. The reverse process with and without inversion error is summarized as

$$\begin{aligned} &\text{(exact reverse)} \quad q_0 \xleftarrow{T_1^{-1}} q_1 \xleftarrow{T_2^{-1}} \cdots \xleftarrow{T_N^{-1}} q_N = q, \\ &\text{(computed reverse)} \quad \tilde{q}_0 \xleftarrow{S_1} \tilde{q}_1 \xleftarrow{S_2} \cdots \xleftarrow{S_N} \tilde{q}_N = q. \end{aligned} \tag{32}$$

The computed transport S_n is not the same as T_n^{-1} but the algorithm aims to make the inversion error small. For our theoretical analysis, we make the following assumption on the error.

Assumption 2 (Inversion error)**.** *For $n = N, \cdots, 1$, the computed reverse transport S_n is non-degenerate, in $L^2(\tilde{q}_n)$, and satisfies that*

$$\|T_n \circ S_n - \mathrm{I_d}\|_{\tilde{q}_n} \leq \varepsilon_{\text{inv}}. \tag{33}$$

In practice, the quantity $\|T_n \circ S_n - \mathrm{I_d}\|_{\tilde{q}_n}^2$ can be empirically estimated by sample average, namely the mean-squared error

$$\begin{aligned} \text{MSE}_{\text{inv}} &= \frac{1}{n_{\text{inv}}} \sum_i \|T_n \circ S_n(x_i) - x_i\|^2, \\ x_i &= S_{n+1} \circ \cdots \circ S_{N-1}(z_i), \quad z_i \sim q, \end{aligned}$$

computed from n_{inv} test samples. With sufficiently large n_{inv}, one can use MSE_{inv} to monitor the inversion error of the reverse process and enhance numerical accuracy when needed. It was